MINISTRY OF EDUCATION AND TRAINING

FPT UNIVERSITY

MASTER THESIS PROPOSAL

Major: Software Engineering

Design and Development of a Tour Concierge Agent for Travel Agencies Using a Multi-Agent and Agentic RAG Framework

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1. Introduction and Background

1.1. Problem Description

In the context of rapid digital transformation in the tourism industry, travel agencies (tour operators) are still facing significant challenges in managing and supporting customers throughout the tour journey. Artificial Intelligence (AI) is increasingly recognized as a key driver in reshaping travel services and customer experience [1], [2]. However, current tour operations still rely heavily on human resources such as customer service staff and tour guides, which leads to several limitations, including repetitive customer inquiries, lack of 24/7 support, limited personalization, and high operational costs [3].

These issues directly affect customer experience, particularly in group tours where travelers require continuous support regarding schedules, meeting points, and unexpected situations. Tour guides are often overloaded with repetitive and minor requests, reducing overall service quality [3].

Despite the growing adoption of AI chatbots and travel assistants, existing systems primarily rely on single-agent architectures and static retrieval mechanisms, which limit their ability to provide context-aware, adaptive, and personalized support during dynamic tour scenarios [4].

In particular, current Retrieval-Augmented Generation (RAG) approaches are typically passive, lacking agent-level reasoning, task decomposition, and memory mechanisms for continuous interaction [5]. Furthermore, there is limited research on applying multi-agent architectures combined with Agentic RAG in real-time tour operations [6].

Therefore, a clear research gap exists in designing a multi-agent Tour Concierge system that integrates planning, retrieval, execution, and personalization capabilities, supported by adaptive memory and data-driven user modeling.

This project aims to contribute both technically and practically by proposing a multi-agent architecture for Tour Concierge systems, including specialized agents such as Planner Agent, Retrieval Agent, and Interaction Agent; designing an Agentic RAG framework that incorporates task decomposition, tool usage, and memory-aware retrieval; introducing a personalization mechanism based on user behavior and tour context; and evaluating the effectiveness of the proposed system against baseline approaches (e.g., single-agent chatbot, standard RAG).

1.2. Research Objectives

General Objective**:** To design and propose a Tour Concierge Agent model that supports customers during tour journeys, improving user experience and operational efficiency for travel agencies.

Specific Objectives (SMART):

- Identify key pain points in the customer journey during tours (within the initial 2–3 weeks)
- Analyze existing methods and technologies in Travel AI Agents (during the literature review phase)
- Design a Tour Concierge Agent model suitable for tour operators
- Develop a prototype or conceptual model (if feasible)
- Evaluate the effectiveness of the proposed solution based on usability, response time, and user satisfaction

Research Questions: To achieve the aforementioned goals, this research will systematically investigate the following questions:

1. What are the existing methods to solve customer support challenges in tour operations? What are their advantages and limitations?
2. What AI-based solution (Tour Concierge Agent) can be proposed, and how will it be designed and implemented?
3. How can the effectiveness of the proposed solution be evaluated in a real-world or simulated environment?

1.3. Scope of the Project

In-Scope (Included within the project)**:**

- Research and design of a Tour Concierge Agent model
- Focus on in-tour support (during the tour)
- Context-aware customer support system design
- Prototype or mockup development (if applicable)
- Evaluation through surveys or simulated scenarios

Out-of-Scope (Explicitly excluded from the project)**:**

- Full-scale production system development
- Deep integration with real enterprise systems
- Advanced AI model training from scratch

2. Proposed Solution

2.1. Solution Description

This project proposes a Tour Concierge Agent, an AI-powered system designed to provide continuous, context-aware, and personalized support for customers during organized tours. Unlike traditional chatbot systems that rely on single-agent architectures and static response mechanisms, the proposed solution adopts a multi-agent architecture combined with an Agentic Retrieval-Augmented Generation (Agentic RAG) framework.

The system is structured as a set of specialized agents, each responsible for a specific function within the overall workflow. A Planner Agent is responsible for interpreting user requests and decomposing them into smaller tasks. A Retrieval Agent handles the process of retrieving relevant information from structured data sources such as tour itineraries, FAQs, and operational guidelines. An Executor (or Response) Agent generates context-aware responses by integrating retrieved knowledge with language model capabilities. In addition, a Memory Module is incorporated to store interaction history and user preferences, enabling the system to maintain context across multi-turn conversations and provide more personalized assistance.

The proposed system leverages an Agentic RAG framework, where information retrieval is dynamically guided by agent reasoning rather than being a passive process. This allows the system to adapt its retrieval strategy based on the user's intent, current tour context, and previous interactions, resulting in more accurate and relevant responses compared to conventional RAG-based chatbots.

To further enhance system intelligence, a data-driven component is integrated into the solution. User interaction logs and feedback data will be analyzed to identify behavioral patterns and common user needs during tours. These insights can be used to refine system responses and support basic personalization strategies. Additionally, structured representations of tour data (e.g., itinerary-based knowledge structures) may be utilized to improve retrieval accuracy and contextual understanding.

Overall, the proposed Tour Concierge Agent aims to provide real-time, context-aware assistance throughout the tour, reduce the workload for tour guides and customer support staff, improve response accuracy and user satisfaction, and demonstrate the effectiveness of combining multi-agent systems with Agentic RAG in a real-world application domain.

This solution not only addresses practical challenges in tour operations but also contributes to the research area of AI agents by exploring the integration of multi-agent coordination, adaptive retrieval, and user-centric personalization.

It is hypothesized that the proposed multi-agent Tour Concierge Agent will achieve higher response accuracy and task completion rates compared to baseline systems, including rule-based chatbots and single-agent RAG models.

Specifically, the system is expected to:

- Improve response accuracy by approximately 15–25% compared to traditional chatbot systems
- Increase task completion rate in tour-related queries
- Maintain response latency within an acceptable threshold (e.g., under 2–3 seconds)
- Achieve higher user satisfaction scores based on survey evaluation

2.2. Software Architecture

The system will follow a **client-server architecture** integrated with an AI layer.

Main components:

- Frontend: Mobile app / Chat interface (e.g., messaging platforms)
- Backend: API server handling business logic
- AI Layer: LLM-based conversational agent
- Data Layer:
    - Tour itinerary database
    - FAQ database
    - Customer data

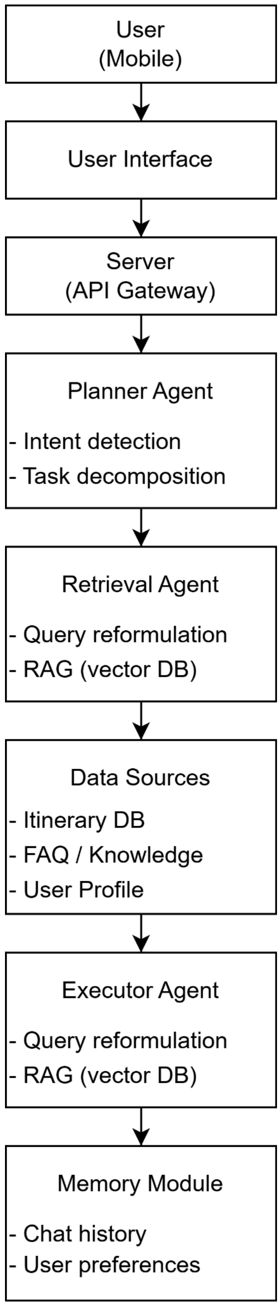

Figure 1: Architectural diagram illustrating the proposed data flow.

2.3. Technology Stack

This project will utilize a combination of modern technologies and development tools to ensure efficient system design, rapid prototyping, and scalability.

Table 1: Technologies Used

Category	Technology	Description
Backend Development	Python (FastAPI / Flask)	Used to build RESTful APIs and handle business logic
Frontend Development	Flutter (Mobile Application)	Cross-platform mobile app interface for user interaction
AI Layer	Large Language Model APIs (LLMs)	Powers conversational capabilities and context-aware responses
AI Enhancement	Retrieval-Augmented Generation (RAG)	Retrieves relevant data to improve response accuracy
Database (Option 1)	MongoDB	NoSQL database for flexible, semi-structured data
Database (Option 2)	PostgreSQL	Relational database for structured data
UI/UX Design	Figma	Design tool for wireframes and interface prototypes
API Testing	Postman	Tool for testing and debugging APIs
Version Control	Git	Source code management system
Code Repository (Optional)	GitHub / GitLab	Platform for hosting and managing source code

3. Implementation Plan

3.1. Main Development Stages

The project will be implemented in 6 stags:

- Stage 1 (Requirements Gathering)**:** Identify the problem context, analyze current tour operation workflows, and define system requirements, use cases, and target users for the Tour Concierge Agent.
- Stage 2 (Literature Review and Analysis)**:** Review existing studies on AI agents, chatbots, and travel-related systems; analyze current solutions and identify research gaps to support the proposed approach.
- Stage 3 (System Design)**:** Design the overall system architecture, including the AI agent workflow, data structure (itinerary, FAQ, user profiles), and user interaction flows.
- Stage 4 (Prototype Development)**:** Develop a functional prototype by implementing backend services, integrating AI APIs, and building a basic frontend interface using simulated or mock data.
- Stage 5 (Testing and Evaluation)**:** Conduct functional and scenario-based testing, evaluate system performance (e.g., response accuracy, usability), and collect user feedback for analysis.
- Stage 6 (Final Report Preparation)**:** Document the entire project, including research findings, system design, implementation, evaluation results, and recommendations for future improvements.

3.2. Expected Schedule

The project is estimated to be completed over a 12-week timeframe. The following schedule outlines the key milestones and major activities in each phase:

- Week 1–2 (Problem Definition and Requirements Gathering): Define the research problem, project scope, objectives, and identify system requirements and use cases.
- Week 3–4 (Literature Review and Analysis): Conduct a review of relevant studies on AI agents, chatbots, and travel systems; analyze existing solutions and identify research gaps.
- Week 5–6 (System Design): Design the overall system architecture, including AI workflow, data structures, and user interaction flows.
- Week 7–9 (Prototype Development): Develop the system prototype, including backend implementation, AI integration, and frontend interface using mock data.
- Week 10–11 (Testing and Evaluation): Perform functional and scenario-based testing; evaluate system performance and collect user feedback.
- Week 12 (Final Report and Presentation Preparation): Complete documentation, finalize the report, and prepare presentation materials for project defense.

3.3. Feasibility Assessment

This project is considered feasible within the scope and timeframe of a capstone project, although several challenges may arise during its development. One key challenge is the limited access to real-world data from travel agencies, which may affect the realism and accuracy of the system. In addition, simulating actual user behavior in a tour context can be difficult, as user interactions are often dynamic and situation-dependent. Time constraints also pose a limitation, particularly when integrating multiple components such as backend services, AI models, and frontend interfaces.

To mitigate these challenges, the project will utilize synthetic or mock data to simulate real-world tour scenarios. User surveys and scenario-based testing will be conducted to approximate realistic user interactions and evaluate system usability. Furthermore, the project scope will be clearly defined and controlled to ensure that development remains manageable within the given timeframe.

In terms of resources, the project requires only standard and accessible tools, including a personal computer for development, access to AI APIs for implementing the conversational agent, and common software development tools such as programming frameworks, version control systems, and design tools. These resources are readily available, making the project technically and practically achievable.

4. Expected Results

Upon completion, the project is expected to deliver several key outcomes that demonstrate both theoretical and practical contributions. First, a well-defined Tour Concierge Agent model will be developed, including its system architecture, core functionalities, and operational framework within the context of travel agencies.

In addition, a functional prototype will be implemented, which may take the form of a mobile application or chat-based interface. This prototype will demonstrate the core features of the system, such as itinerary support, contextual question answering, and basic user interaction using simulated data.

The project will also produce **c**ollected and analyzed data from user surveys or scenario-based evaluations. This data will provide insights into user needs, system usability, and overall acceptance of the proposed solution.

Finally, an evaluation report will be presented to assess the effectiveness of the system based on predefined criteria such as usability, response accuracy, and user satisfaction. The results will help validate the feasibility of applying a Tour Concierge Agent in real-world travel agency operations.

5. Evaluation Plan

The proposed system will be evaluated using a combination of quantitative metrics and qualitative assessments to measure both system performance and user experience. The evaluation aims to validate the effectiveness of the proposed multi-agent Tour Concierge Agent with Agentic RAG compared to baseline approaches.

Evaluation Metrics:
The system will be assessed using the following key metrics:

- Response Accuracy: Measured based on the relevance and correctness of responses, using human evaluation or scoring methods (e.g., relevance rating on a Likert scale).
- Task Completion Rate: The percentage of user requests that are successfully resolved by the system.
- Response Latency: The average time taken to generate responses, ensuring real-time usability (target: under 2–3 seconds).
- User Satisfaction: Measured using standardized surveys such as Likert scale or System Usability Scale (SUS).
- Context Retention / Consistency: The ability of the system to maintain context across multi-turn interactions.

Baseline Comparison:
To demonstrate the effectiveness of the proposed approach, the system will be compared against the following baseline models:

- A rule-based or traditional chatbot system
- A single-agent LLM system without RAG
- A standard RAG-based system without multi-agent coordination

This comparison will highlight the impact of both multi-agent architecture and Agentic RAG on system performance.

Evaluation Methods:
The evaluation will be conducted through:

- Scenario-based testing: Simulated tour scenarios (e.g., schedule inquiries, unexpected changes) to assess system behavior in realistic conditions.
- User surveys: Collecting feedback on usability, satisfaction, and perceived usefulness.
- Quantitative analysis: Measuring and comparing performance metrics (accuracy, completion rate, latency) across different system configurations.

Hypothesis Testing:
It is hypothesized that the proposed system will outperform baseline models by:

- Achieving higher response accuracy (expected improvement: ~15–25%)
- Increasing task completion rate in tour-related queries
- Providing better user satisfaction due to context awareness and personalization

The evaluation results will be analyzed to validate these hypotheses and identify areas for further improvement.

6. References

[1] McKinsey & Company, “The Future of Travel: AI and Digital Transformation,” 2023.

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[5] OpenAI, “GPT-4 Technical Report,” 2023.

[6] J. Wu et al., “AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation,” 2023.